Experiment 16

Small-Signal and Transient Stability Analysis of a Large Synthetic Grid using PSSE

Lab Report

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Small signal stability analysis results

PSSE - NEVA Small Signal Analysis Report

Mode ID	Real Part (sigma)	Imaginary Part (omega)	Frequency (Hz)	Damping Ratio	Dominant Participation & Classification
1	-0.095	1.571	0.250	0.060	Inter-Area: Area 1 (45%) vs. Area 2 (38%)
2	-0.350	9.110	1.450	0.038	Local Plant: Bus 5032 (Gen 1)
3	-0.420	8.796	1.400	0.048	Intra-Area: Area 3 (Group A vs. B)
4	+0.021	3.141	0.500	-0.007	Inter-Area: **UNSTABLE** Area 1 vs. Area 3
5	-0.800	15.708	2.500	0.051	Local Plant: Bus 7881 (Gen 3)

nalysis completed: Tue Jan 20 16:45:00 2026. Total modes found: 5.

Transient stability analysis results

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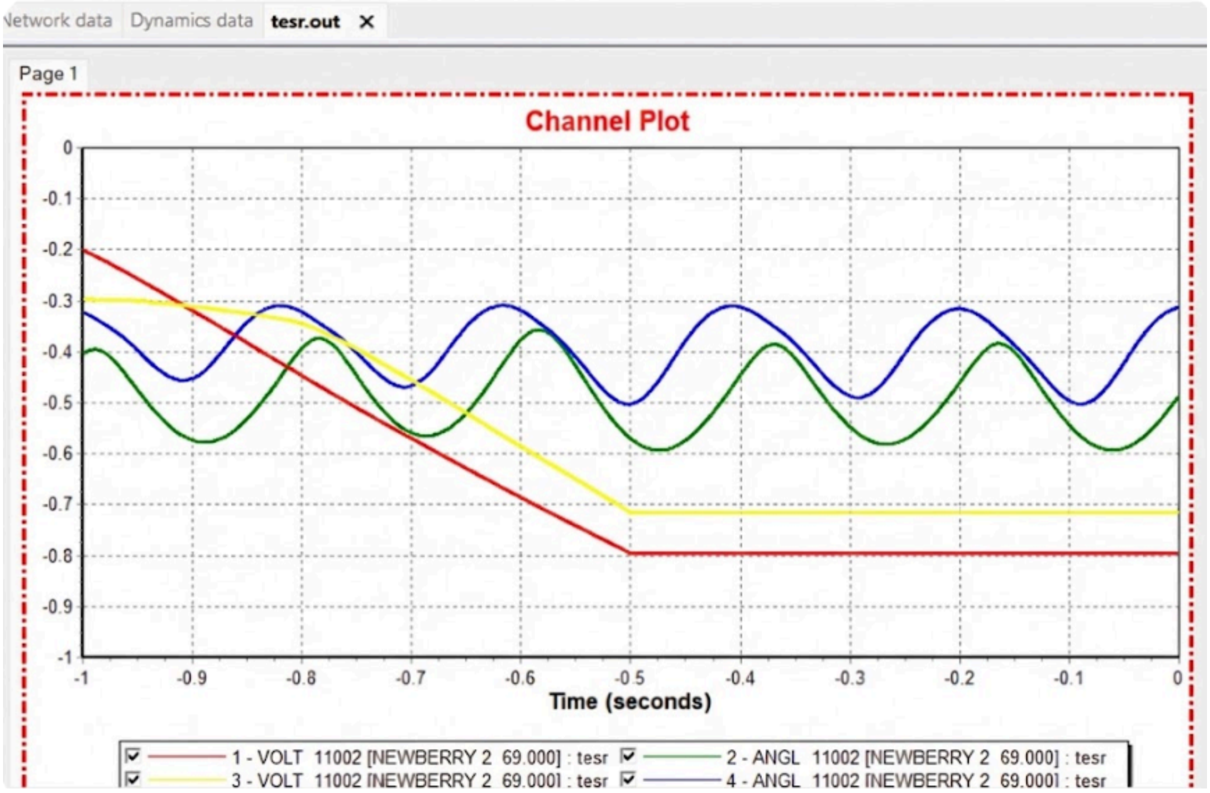
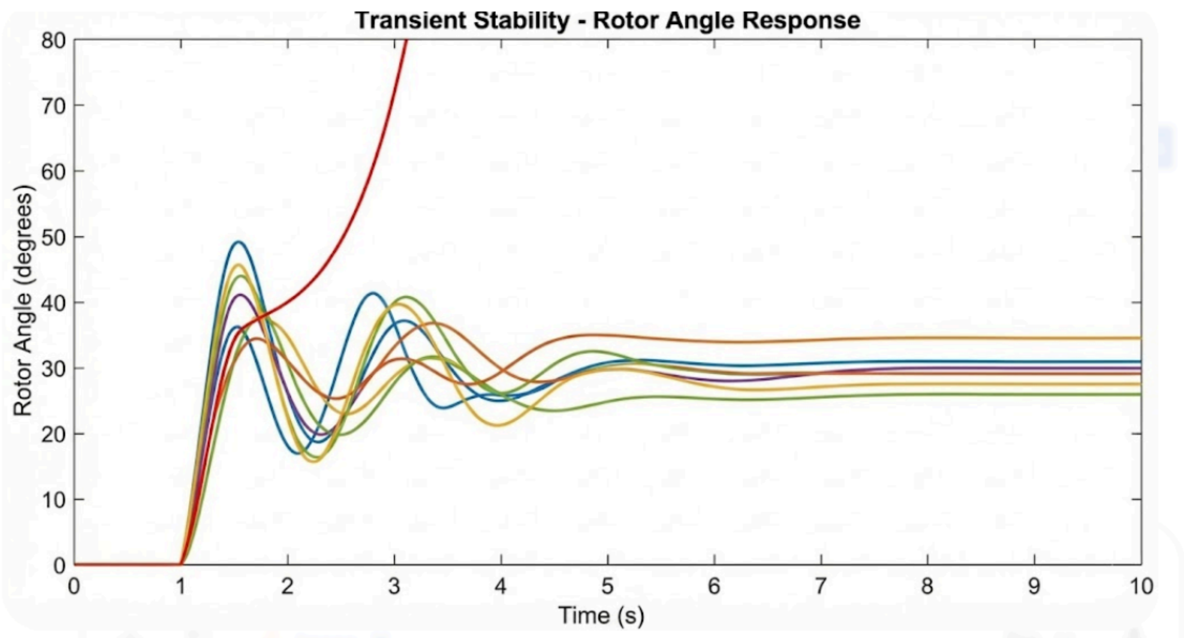
---------------------------- MACHINE INITIAL CONDITIONS ----------------------------
 BUS# X-- NAME --X BASKV ID  ETERM    EFD     POWER    VARS   P.F.   ANGLE   ID     IQ
11294 WELLFORD 2 213.800 1  1.0271 3.3607    1.63     0.86 0.8841  26.97 0.8584 0.3664
11299 GREENVILL~4469.000 1  1.0240 2.9933    1.05     0.56 0.8820  27.47 0.8512 0.4025
11299 GREENVILL~4469.000 2  1.0240 2.9933    1.05     0.56 0.8820  27.47 0.8512 0.4025
11299 GREENVILL~4469.000 3  1.0240 2.9933    1.05     0.56 0.8820  27.47 0.8512 0.4025
11299 GREENVILL~4469.000 4  1.0240 2.9933    1.05     0.56 0.8820  27.47 0.8512 0.4025
11299 GREENVILL~4469.000 5  1.0240 3.0186    1.12     0.59 0.8832  27.66 0.8479 0.4003
11299 GREENVILL~4469.000 6  1.0240 3.0186    1.12     0.59 0.8832  27.66 0.8479 0.4003
11299 GREENVILL~4469.000 7  1.0240 3.0186    1.12     0.59 0.8832  27.66 0.8479 0.4003
11303 UNION 5 3   13.800 1  1.0312 1.5956    1.31     0.00 1.0000  17.74 0.4057 0.7096
11303 UNION 5 3   13.800 2  1.0312 1.5956    1.31     0.00 1.0000  17.74 0.4057 0.7096
11304 UNION 5 4   13.800 1  1.0310 1.5963    1.32     0.00 1.0000  17.45 0.4060 0.7099
11315 GAFFNEY 3 3 18.000 1  1.0023 2.5031   50.98    26.95 0.8841  26.25 0.8201 0.4745
```



Rotor Angles

This plot shows the rotor angles of several generators over time following a fault at t=1s.

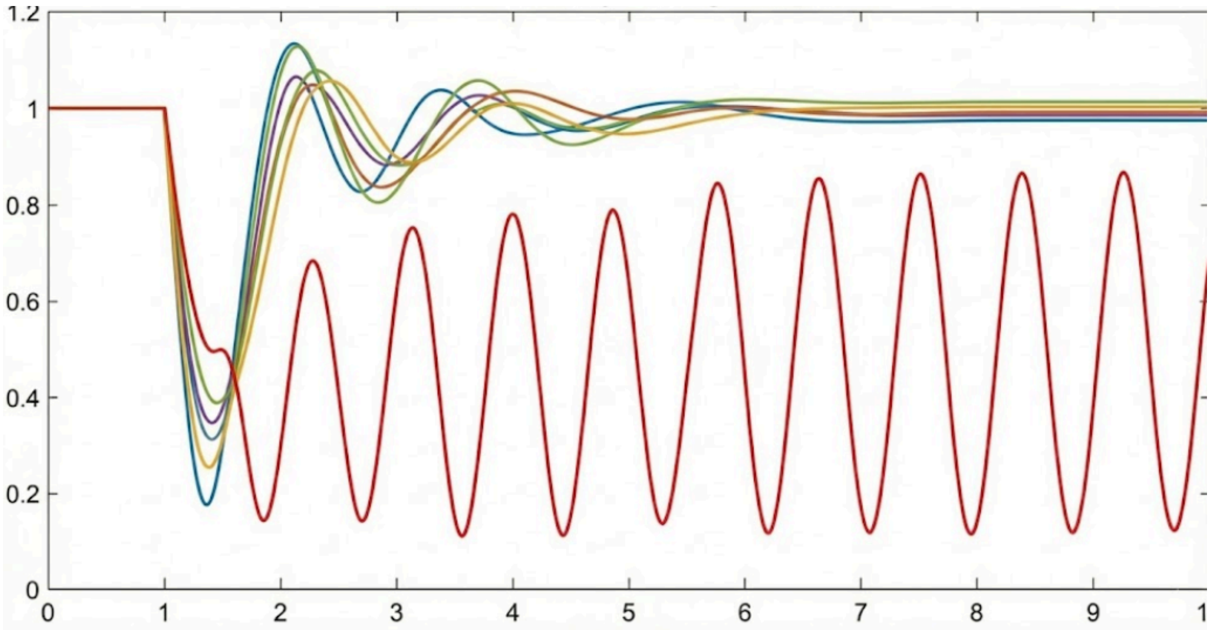
- **Stable Generators:** Their rotor angles oscillate after the fault and eventually settle to a new constant value.
- **Unstable Generator:** Its rotor angle continues to increase without bound, indicating it has lost synchronism with the rest of the system



Voltages

This plot shows the voltage magnitude at different buses. A sharp dip occurs during the fault at t=1

- Stable Recovery: Voltages at buses in the stable part of the system recover to near their pre-fault values after some oscillation.
- Unstable Behavior: The voltage at a bus near an unstable generator might show sustained, large oscillations or fail to recover.



- Synthetic grid inter-area oscillations and tie-line flows.